Sr.No. 8385

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26 | Session: Odd MSE October 2025 | Date: 08-10-2025
Sem: 1 | Slot: 10:00am - 11:00am | C. Code: CS413
Exam Type: Written | Block:305 Course Section:1 | Seat No: 2022700022

1_41_634_986_1_1505_41450

Name of the candidate. Kaumod Bagave

Candidate's UID number: | 2 | 0 | 2 | 2 | 7 | 0 | 0 | 0 | 2 | 2 |

Examination class room number: | 3 | 0 | 5 |

Invigilator's signature with date: 8/10

(To be filled in by the candidate)

EXAMINATION: Btech MSE
(For example FY MCA/FY BTECH)

PROGRAM: CS (DS)

DAY AND DATE: 8th October, 2025

COURSE NAME: Deep Learning

SEMESTER : ~~I / II / III / IV / V / VI~~ / VII / ~~VIII~~

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

8328

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
3) Do not write anything in the column provided for the marks to be assigned by the examiners.
4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
12) The answer-books will be scrutinized before they are sent to examiners.
13) All answer-books supplied shall be returned whether written or blank.
14) Nothing shall be written on the question-paper.
15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q1.	
	a)	$J(\omega) = \frac{1}{2} \lVert X\omega - y \rVert^2$
		To prove $\eta < \frac{2}{\lambda_{max}}$ where λ_{max} is the largest eigenvalue of $X^T X$

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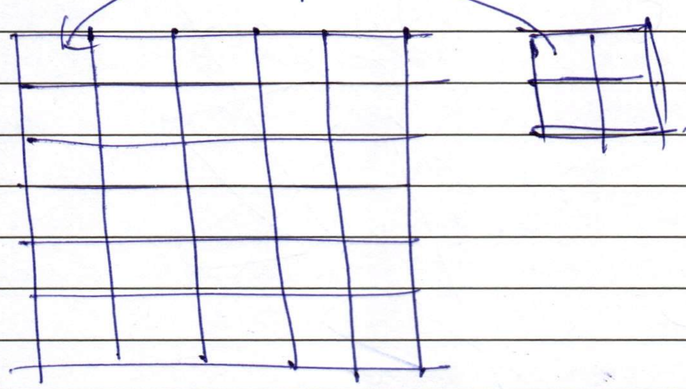
	Q.2]	
	a)	number of multiplications for 2D convolution

let ~~Ha~~ H and W be height and width of the input respectively
and $K \times K$ be the kernel.

standard ~~comp~~ convolution Stride = 0



First component, every grid square excluding the outermost K squares is passed horizontally by the ~~kernel~~ entire kernel once.

This means that for $(W - K)$ squares they are ~~passed~~ multiplied K times.

$(W-2K)K$:- 1

2nd

In the vertical direction, every grid square except the outermost K are passed vertically by the entire kernel once.

for $(H-2K)$ squares they are multiplied K times

$(H-2K)K$

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for the outermost, the each of them are passed only once by the filter.

$2HK + 2WK$.

$$\text{total} = (H-2k)k + (W-2k)k + 2HK + 2WK$$

for a 3D depthwise convolution of input size

$H \times W \times C$ and K filters of size $F \times F$,

we can apply similar logic to get.

$$(H-2F)F + (W-2F)F + (C-2F)K$$

$$+ \quad 2HF + 2WF + 2CK$$

Depthwise seperable convolutions is a form of 3D convolution which employs a 2D filter kernel over a 3D grid of inputs. The filter convolutes the first layer at the first depth and ~~an~~ then on ~~ano~~ the next and so on.

It is quite important because it provides layered convolution over a 3D space while using 2D filters.

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Q.2]

b) undercomplete auto-encoders are autoencoders where the ~~tot~~ number of neurons in the input space is greater than the number of neurons in the latent space.

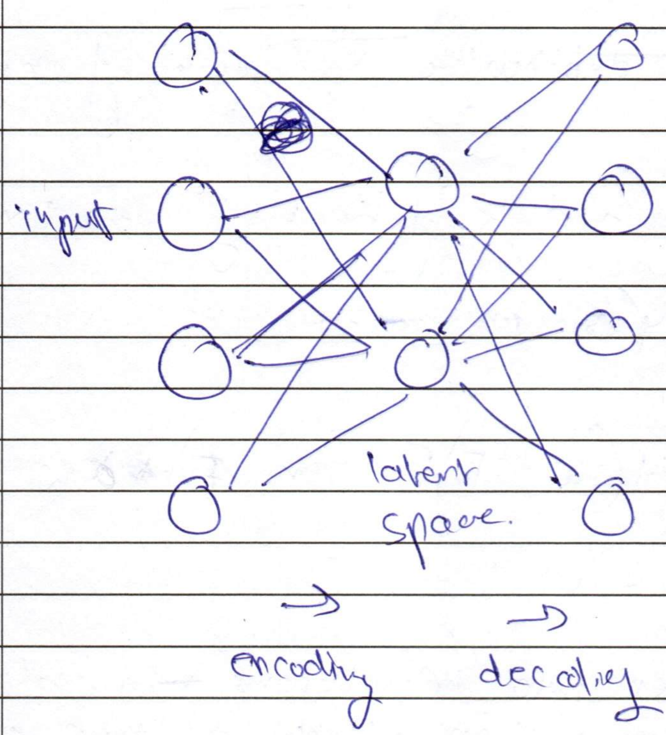

Fig. under-complete auto encoder.

This achieves dimensionality reduction as the autoencoder learns the important components of the input and ~~genero~~ generates the output based on these.

PCA (Principle component analysis) is also a dimensionality reduction technique that uses a covariation relationship intermediate and eigen vectors to calculate principle component values for each instance.

The covariation matrix calculation resembles the MSE training on the autoencoder and they are equivalent.

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An over-complet autoencoder is an auto-encoder that has a larger amount of latent space neurons than input space neurons.

Regularization is a technique used on over-complete autoencoders to ~~minimize the~~ maximize the learning potential of an autoencoder.

However, without Regularization, the input components are directly projected onto the latent space. This results in lossless and exact encoding of the input onto the latent space. with spare neurons it also learn every minute noise and detail in the input and after decoding, the output is calculated exactly as the input, which is called an identity function.

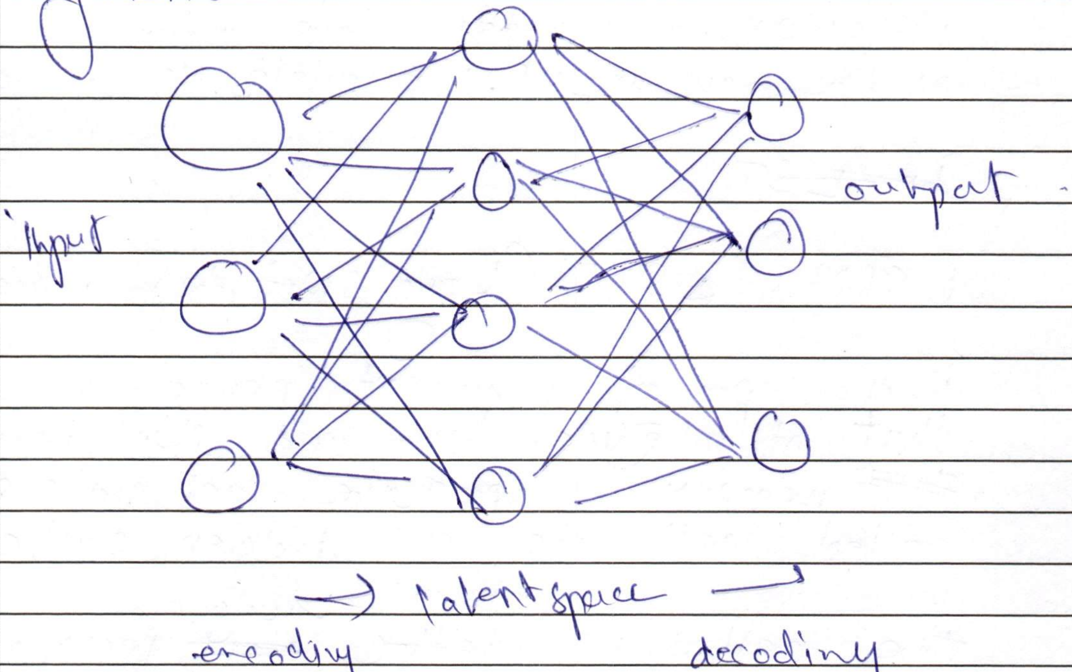

Fig. An overcomplete autoencoder.

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Q3]

a))

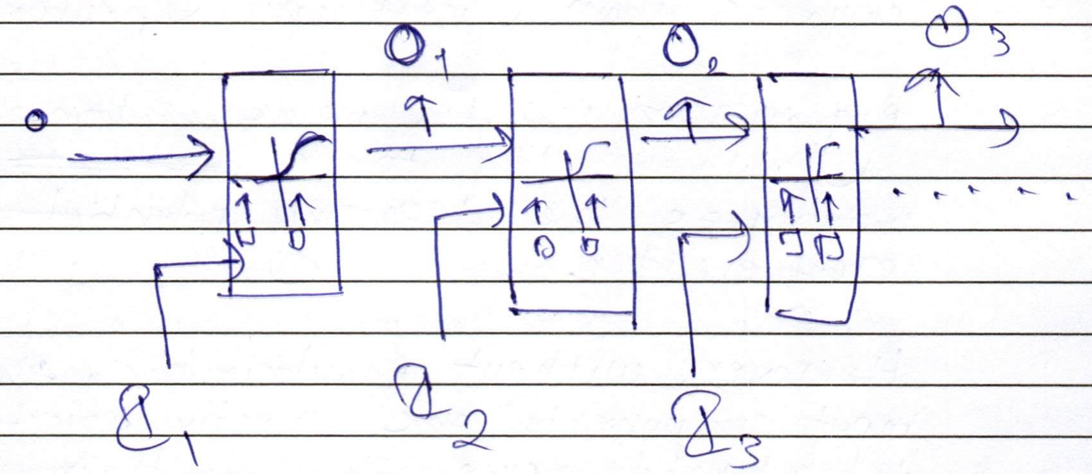

Fig. A recurrent neural network (RNN)

A Recurrent neural network is a neural network that can work on sequential data streams that takes the input from the previous iteration as a hidden state input as well as the new input, to calculate the next output.

~~LST~~:

LSTM: LSTM (Long short term memory) is a type of recurrent neural network that introduces long and short term ~~that~~ memory as separate cell state streams called cell state and hidden state.

the cell state ~~is the~~ acts as ~~short~~ long term memory and hidden state is the short term memory.

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It has 3 gates as components : input gate, output gate, and forget gate.

Let f, i, o be the intermidiate calculations of forget, input and output states respectively

~~let~~ h_t be the ~~current~~ hidden state value at t

C_t be the cell state value at t.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\bar{C}_t = (C_{t-1} \times f_t) + (i_t \times \tanh(h_{t-1}, x))$$

$$\cancel{C_t = \bar{C}_t \times \tanh(C_t)}.$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$h_t = \tanh(C_t) \times o_t$$

The vanishing and exploding gradient problem is a problem pertaining to RNNs where the repeated recursion of the same weights ~~in~~ of the hidden state increases or decreases the gradient to an infeasible value.

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~~as~~ the sequence goes on, ~~it~~ if weights are below 1, the gradient vanishes over time, and if above 1, the gradient increases exponentially and becomes too large or explodes.

The LSTM architecture prevents this from happening as the forget gate exists.

The forget gate basically decides what percentage of the existing long term memory to take into account based on the current short term memory and inputs. This prevents the gradient from ballooning into something too large or shrinking into something too insignificant, solving the vanishing gradient problem.

Q.3)

b)

→ GRU (gated recurrent unit) is a type of neural network that is built upon the LSTM architecture that aims to reduce complexity and parameters.

A gru has two gates, a forget gate, and an update gate, these two in combination maintain the constant cell state stream that is equivalent to a combination of long and short term memory.

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